

Elicit vs. Teach, Mechanistically: Eleven Metrics on the TinyStories-1B Twins

MARS V — Mechanistic Understanding of Elicitation vs. Teaching

Working results document, September 2026

Setup

The experiment. Two identical copies (“twins”) of a 1B-parameter language model pretrained only on children’s stories — no arithmetic anywhere in the corpus — are fine-tuned on the same target task: natural-language addition and subtraction of 1–4-digit numbers, **What is the difference between 8 and 5784? $\rightarrow$ -5776.** The training protocol is identical for both. The *only* difference is the checkpoint the fine-tune starts from:

- **Teach twin:** the blank base model. The capability is absent.
- **Elicit twin** (“TS1B-latent”): the blank base plus two installations, neither of which shows the target task: (i) arithmetic in symbols only ($23 + 45 = 68$, 1M examples, full fine-tuning) and (ii) a word $\leftrightarrow$ symbol binding dose that only *rewrites* questions (**What is the sum of 621 and 5068? $\leftrightarrow$ 621 + 5068**) and never shows an answer. This twin can compute in symbols and knows what the words mean, but on the target task it scores 0.000: asked a question, it rewrites it instead of answering. The capability is latent.

Every construction step has a control (Table 1); the two twins are behaviourally identical on the target task before fine-tuning.

Construction step / check	value	control
symbol install: exact match on $23 + 45 =$	0.67–0.73	blank twin: 0.00
corpus census: “sum” in 439M words of stories	$21\times$	“equals” 17, “minus” 57
binding dose: question $\rightarrow$ symbols rewrite exact	0.484	blank + same dose: 0.000
sum-only dose on <i>difference</i> questions	writes “+”	binding is per word
self-chain: rewrite the question, then feed the model its own rewrite	0.328 ($= 0.484 \times 0.668$)	blank + same doses: 0.000
exact match on the target task, before fine-tuning	0.000	= blank twin

Table 1: The elicit twin before fine-tuning: can compute in symbols, understands the words, still answers nothing — and the two parts compose (self-chain = rewrite accuracy $\times$ symbol accuracy, a product that held on four variants). Everything below is about what differs inside the two twins and what that difference does.

The pair every metric is reported on. The teach parent used throughout is the *format-installed* twin: the blank base first fine-tuned on a random-label dose in the target’s own format, so that it already emits a number after every question (format validity 1.000) without ever having seen a correct one (exact match 0.000). This is the paper’s “pre-teach format” control and makes the two parents design-matched: both are the blank base plus one full-fine-tuning dose, both already output numbers, only one has the algorithm. Both parents are then fully fine-tuned with the identical recipe on the identical 4M-example file to convergence: **elicit FT** reaches exact match 0.986 in 21,500 steps; **teach FT** reaches 0.424 in 58,500 steps (Table 2). Teaching this base

needs full fine-tuning: under the paper’s adapter protocol (LoRA) the taught twin stops by its own convergence rule at 0.09–0.14, where the paper’s Figure 2 puts it. Other endpoints appear only where they add something: the LoRA runs carry the learning-curve sweeps and the training snapshots (M1, M5, M7), and the teach twin fine-tuned from the *blank* base (0.819) is the control that separates the algorithm from the format (M8) and the robustness check in Section E.

endpoint	parent	method	data	steps (passes)	stop	exact match
elicit FT	TS1B-latent	full FT	4M	21,500 (0.7)	converged	0.986
teach FT	format-installed	full FT	4M	58,500 (1.9)	converged	0.424
elicit LoRA (sweep, snapshots)	TS1B-latent	LoRA $r=512$	1M	11,500 (1.5)	converged	0.981
teach LoRA (sweep)	format-installed	LoRA $r=512$	1M	–	converged	0.079
teach LoRA (snapshots)	blank	LoRA $r=512$	1M	25,000 (3.2)	converged	0.093
teach FT from the blank base (control)	blank	full FT	4M	72,000 (2.3)	converged	0.819

Table 2: The main pair (top) and the endpoints used where noted (bottom). LoRA learning rate 3.53×10^{-4} , full FT 2×10^{-5} ; batch 128; seed 316; training stops when validation loss stops improving (eps/k rule). Exact match = 0-shot accuracy on 1,024 held-out problems. The 4M file is the 1M file plus 3M new unique problems, disjoint from every evaluation set.

Two terms used throughout. *Answer position*: the last prompt token, where the model predicts the first token of the answer; every activation measurement is taken there unless stated. *Residual stream*: the model’s running internal state at a token position, a 2048-dimensional vector that every layer reads and adds to. Everything else is defined where it is used.

How to read this document. Eleven metrics, each with the same three blocks: **Elicit vs. teach, in one sentence** (what the metric asks and which way each regime should fall), **How it is measured** (the procedure, complete enough to reproduce), **The numbers** (elicit FT vs teach FT; every value is in the table or figure it points to; other endpoints only where they add a control or an intuition). The metrics are grouped by what they look at: the *cost* of learning (M1), the *machinery* that results (M2–M4), the *dynamics* of building it (M5–M8), and the *latent state* before any target training (M9–M11). Table 11 at the end puts every metric for the pair on one page, with the robustness endpoints beside it.

A. Cost of learning

M1 Learning cost — *cheaper with every example vs. the hump*

Elicit vs. teach, in one sentence. Does each new example cost less than the last, because it only connects a capability that exists (elicit), or does the cost first rise, because structure is being built (teach)?

How it is measured. Prequential excess description length (EDL), the paper’s learning-cost measure. Train on a stream of n examples; before each example is trained on, record the extra nats the model needs to predict its answer tokens beyond a converged reference; sum over the stream and divide by the number of answer tokens (EDL per token). Repeat on a sweep $n = 100 \dots 4 \times 10^6$ with the identical LoRA protocol (19 rungs per arm, all converged), plus 0-shot exact match on 1,024 held-out problems at each rung. A falling curve means each example costs less than the last; a down-up-down curve (“the hump”) means mid-run examples are the most informative because structure is being built. Two arms: the elicit twin and the format-installed teach twin; the blank teach twin’s sweep is shown alongside to make

the format’s share of the cost visible. The full-FT endpoints have no sweep; their cost is reported as steps to convergence and held-out test loss.

The numbers. Figure 1, Table 3. EDL/token falls monotonically for the elicit twin ($4.53 \rightarrow 0.09$) and goes down-up-down for the teach twin (minimum 0.64 at 10K, peak 1.33 at 100K, 0.88 at 1M). Exact match at 1M: 0.98 vs 0.08; under full FT 0.986 vs 0.424, in 0.7 vs 1.9 passes. The blank teach twin has the same hump, higher (0.78 / 2.02 / 1.40): the format is a fixed cost on top, not the hump. *Reading:* eliciting connects a capability that exists; teaching builds one, and the hump is the algorithm being built.

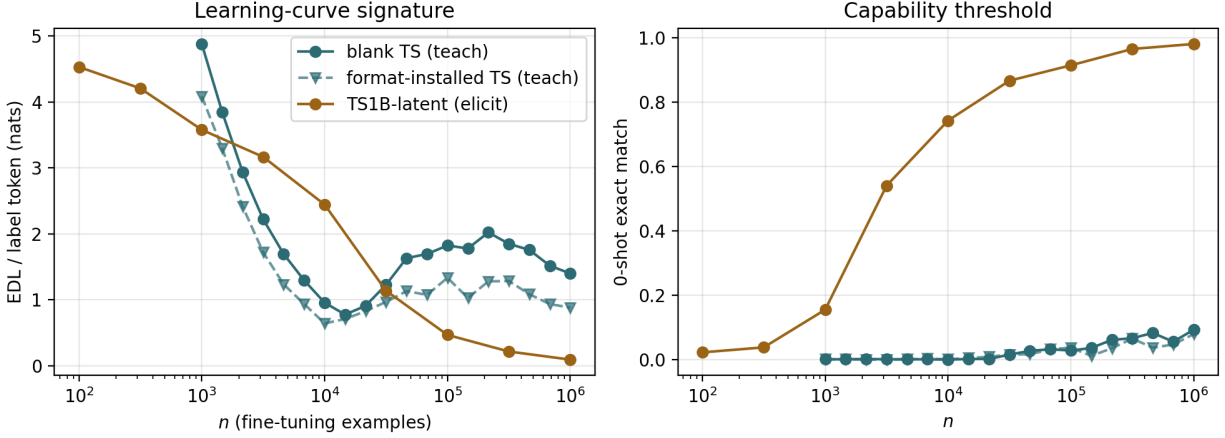


Figure 1: EDL/token (left) and 0-shot exact match (right) vs n ; teal = teach twin from the blank base, dashed teal = teach twin from the format-installed base, gold = elicit twin.

n	elicit (TS1B-latent)		teach (blank)		teach (format-installed)	
	EDL/tok	EM	EDL/tok	EM	EDL/tok	EM
100	4.533	0.023	6.511	0.007	—	—
316	4.208	0.038	6.387	0.033	—	—
1,000	3.579	0.155	4.882	0.001	4.087	0.001
3,162	3.168	0.541	2.221	0.001	1.720	0.000
10,000	2.444	0.743	0.956	0.000	<i>0.641</i> (min)	0.001
14,678	—	—	<i>0.775</i> (min)	0.002	0.708	0.005
31,623	1.137	0.867	1.228	0.016	0.968	0.016
100,000	0.467	0.915	1.824	0.028	<i>1.333</i> (peak)	0.037
215,443	—	—	<i>2.022</i> (peak)	0.062	1.279	0.035
316,228	0.215	0.966	1.849	0.066	1.285	0.065
1,000,000	0.092	0.981	1.399	0.093	0.880	0.079
4,000,000	—	—	0.930	0.139	—	—

Table 3: Per-rung values, LoRA protocol (EDL in nats per answer token; EM = exact match; 19 rungs per teach arm, all converged; rungs shared with the elicit sweep shown plus each arm’s minimum and peak).

B. The machinery that results

M2 Circuit overlap — *the same circuit vs. a new circuit*

Elicit vs. teach, in one sentence. Does the fine-tuned model solve the task with the circuit its parent already had (elicit) or with a circuit that did not exist before (teach)?

How it is measured. *Building a circuit.* The model is split into 528 nodes: 512 attention heads (32 per layer $\times$ 16 layers; heads move information between token positions) and 16 MLP blocks (one per layer; they transform it in place). For each node, attribution patching scores how much the answer’s logit would change if that node’s activation at the answer position were replaced by its value on a different problem: gradient of the answer logit-difference with respect to the node’s activation, times the activation difference between the clean and the corrupted problem, averaged over 128–256 problem pairs. The 32 highest-scoring nodes are the circuit. A model that cannot do the task gives a noise map, which the tool flags (its answer logit-difference is ≈ 0). *Comparing circuits.* $\text{Jaccard@32} = \text{shared nodes} \div \text{union of the two 32-node sets}$. Two random circuits overlap at 0.031. The *ceiling* is the overlap of two circuits of the *same* model computed from two disjoint halves of the data — how repeatable a circuit is (0.42–0.68 here); an overlap at the ceiling means “the same circuit”. Spearman = rank correlation of the node scores over the union. Also which layers and node types each circuit’s top nodes come from.

The numbers. Table 4. Elicited vs taught circuit: 0.306 against ceilings of 0.56 and 0.52 (score correlation 0.03). The elicited circuit is the parent’s own symbol circuit re-weighted: parent $\leftrightarrow$ elicited child 0.46–0.52, the top four nodes identical and in the same order (mlp:15,14,13,12); and it is the same circuit whichever way it is trained (LoRA $\leftrightarrow$ full FT 0.641, at the ceiling). The teach parent has no circuit to inherit (its map is noise: answer logit-diff 0.0). What the two children share is the late-layer MLP engine (mlp:15,14,13,7); the taught circuit adds early and middle MLPs (mlp:0,1,3,4,5,6,10,11) and attention heads in layers 4–7 and 15 of its own, the elicited one uses mlp:8,9 and three layer-7 heads. *Reading:* elicitation re-weights the circuit the parent already had; teaching builds a related but different one.

Comparison	Jaccard@32	chance	ceiling(s)
elicited child (FT) $\leftrightarrow$ taught child (FT)	0.306 (Spearman 0.03)	0.031	0.561 / 0.524
elicit parent’s symbol circuit $\leftrightarrow$ elicited child	0.455–0.524	0.031	0.684
teach parent $\leftrightarrow$ taught child	undefined (parent map is noise)	–	–
elicited child, full FT $\leftrightarrow$ elicited child, LoRA	0.641 (Spearman 0.54)	0.031	0.561 / 0.684
taught child (FT) $\leftrightarrow$ taught child from the blank base (FT)	0.362 (Spearman 0.18)	0.031	0.524 / 0.422

Table 4: Circuit overlaps: the pair (top) and the references that place it (bottom): what the elicited circuit inherits, that the teach parent has nothing to inherit, that the elicited circuit is method-independent, and that two taught circuits differ from each other too. All maps pass the performing-regime check.

M3 Circuit faithfulness — *does the circuit carry the computation?*

Elicit vs. teach, in one sentence. Do the 32 nodes called “the circuit” really produce the answer — in both regimes — so that every overlap in M2 compares two real mechanisms?

How it is measured. True activation patching, in two directions. *Sufficiency*: run a corrupted problem, copy the clean problem’s activations into only the top- k nodes, and measure what fraction of the clean-vs-corrupt answer logit-difference is recovered. *Necessity*: run the clean problem, corrupt only the top- k nodes, and measure what fraction is destroyed. 128 problem pairs, $k = 8, 16, 32, 64, 128, 528$; 1.0 means the top- k nodes alone account for the whole behaviour.

The numbers. Top-8 of 528 nodes recover / destroy the answer at 0.923 / 0.943 (elicit) and 0.983 / 0.988 (teach); top-32 ≥ 0.996 for both (Table 11). *Reading*: no difference — both regimes build compact, load-bearing circuits, so M2, M4 and M5 compare two real mechanisms.

M4 Head roles — the same reader heads vs. rediscovered ones

Elicit vs. teach, in one sentence. Does the fine-tuned model keep the parent’s heads that read the two numbers (elicit) or have to find reader heads of its own (teach)?

How it is measured. Desiderata Component Masking (Prakash et al. 2024). For one piece of information at a time, take a problem and a counterfactual problem that differs only in that piece, and learn (by gradient descent on a mask over heads, with a sparsity penalty) the smallest set of attention heads such that copying those heads’ activations from the counterfactual run into the clean run makes the model answer the counterfactual problem. That set is the heads carrying that piece (its “role”). The set is validated by its counterfactual-flip accuracy against the ceiling obtained by copying every head; a set is “compact” only if it reaches that ceiling. Role sets are compared across models by Jaccard (chance ≈ 0.05); 64–128 counterfactual pairs, heads only.

The numbers. Table 5. Both children have compact operand-reading sets at their ceilings. The elicited child’s are the parent’s: 0.69 / 0.86 overlap with the engine’s sets (first / second operand) under full FT, 0.88 / 0.90 under LoRA, all at ceiling flip accuracy. The taught child’s sets are larger (32 / 41 heads vs 30 / 27) and contain most of the same heads: 0.88 / 0.66 vs the engine. A taught model that did not learn the task (LoRA, exact match 0.14) has no compact set at all (flip accuracy 0.01–0.03 against a ceiling of 0.14–0.20). *Reading*: operand reading is a layer-0 function of the substrate; elicitation keeps it intact, teaching has to find it again and lands on a larger set — a graded difference, not an absence.

Comparison (first operand / second operand)	$ A $	$ B $	Jaccard	flip acc. / ceiling of A
elicited child (FT) $\leftrightarrow$ engine	31 / 25	30 / 27	0.694 / 0.857	0.98 / 0.98 (at ceiling)
taught child (FT) $\leftrightarrow$ engine	32 / 41	30 / 27	0.879 / 0.659	0.59 / 0.54 (0.59 / 0.55)
taught child (FT) $\leftrightarrow$ elicited child (LoRA)	32 / 41	36 / 30	0.659 / 0.690	–
engine $\leftrightarrow$ elicited child (LoRA), symbol surface	30 / 27	32 / 30	0.879 / 0.900	at ceiling
engine $\leftrightarrow$ elicited child (LoRA), operation role	25	24	0.750	at ceiling
elicited child (FT) $\leftrightarrow$ elicited child (LoRA)	31 / 25	36 / 30	0.811 / 0.833	0.98 / 0.98
taught child that did not learn (LoRA 4M) $\leftrightarrow$ elicited child (LoRA)	12 / 21	36 / 30	0.043 / 0.041	0.01 / 0.03 (0.20 / 0.14)

Table 5: Head roles by Desiderata Component Masking. “Engine” = the elicit parent on the symbol surface, the reference role sets. $|A|$, $|B|$ = size of the two role sets. Bottom: the references (the LoRA-elicited child keeps the engine’s roles almost exactly; a taught model that never learned the task has no roles to compare).

C. Dynamics: how the machinery comes into being

M5 Circuit formation time — *present at step 1 vs. assembled mid-run*

Elicit vs. teach, in one sentence. Is the final circuit present from the first step (elicit) or assembled somewhere in the middle of training (teach)?

How it is measured. Adapter snapshots were saved along the LoRA runs (128 per run, log-spaced); the full-FT runs have no snapshots, so this metric is read on the LoRA pair (elicit twin vs blank teach twin, 1M examples each). For each snapshot, build the circuit exactly as in M2 and take its Jaccard@32 with the run’s final circuit; read against chance (0.031) and the split-half ceiling. The step at which the curve leaves the noise floor is when the circuit starts to exist; the step at which it reaches the ceiling is when it stops changing.

The numbers. Figure 2. Elicit: 0.49 overlap with the final circuit at step 1 ($16\times$ chance), locked from step 1.5K of 11.5K. Teach: at the noise floor until step 1.3K, then built over steps 1.3K–5.4K, where its EDL peaks (M1). *Reading:* the elicited circuit is present before training starts; the taught circuit is assembled mid-run, exactly while the learning cost is rising.

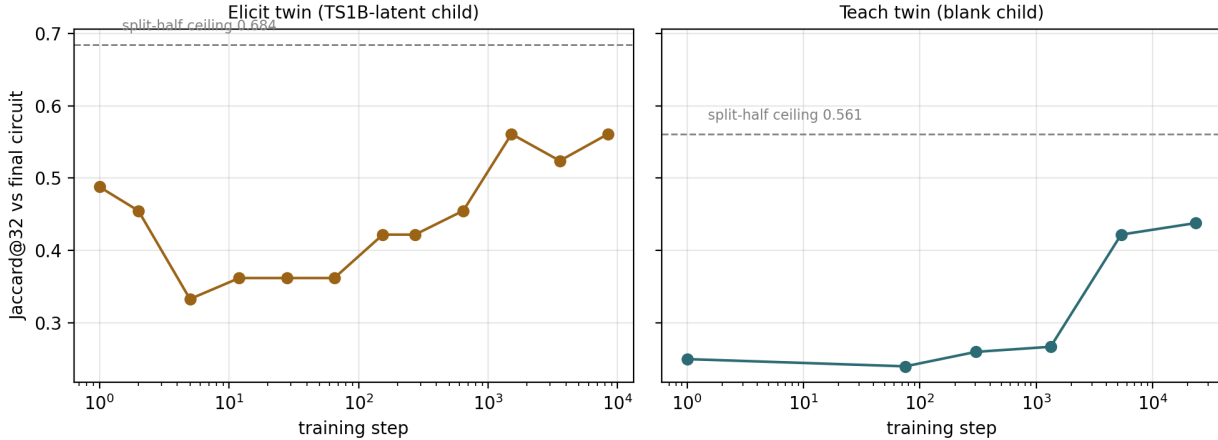


Figure 2: Circuit formation, elicited (left) vs taught (right); dashed = split-half ceiling.

M6 Gradient pressure — *a dying gradient vs. a growing one*

Elicit vs. teach, in one sentence. Does the gradient fade as training proceeds because there is nothing left to change (elicit), or grow because structure is still being built (teach)?

How it is measured. The global gradient norm before clipping, logged at every update. Reported: its peak and the step of the peak; its mean over the first 1%, the middle, and the last 10% of steps; the ratio first-1% $\div$ last-10% (above 1 = decays, below 1 = grows); and the cumulative gradient mass $\sum_t \|g_t\|$. For the LoRA runs also how that mass splits across attention query/key weights (QK), attention value/output weights (VO) and MLP weights; the full-FT trainer logs the global norm only. Norms are comparable within a method, not across methods (different parameter sets).

The numbers. Table 6, Figure 3. Elicit: the gradient peaks at step 1 (34.0) and decays $3.4\times$ from the first 1% of steps to the last 10%. Teach: it starts lower (2.7), peaks at step 50,860 (70.9) and ends $1.4\times$ above its start. Cumulative gradient mass 44K vs 378K ($8.6\times$). Under

LoRA the same shape with a larger gap — elicit decays $3.3\times$, teach grows $40\times$, mass $150\times$ — and the LoRA logs show where the teach gradient goes: attention value/output weights (share 0.87), the operand-fetching heads of M4 being written. *Reading:* elicitation converges and its gradient dies; teaching keeps generating pressure while the structure is built.

	elicit FT	teach FT	elicit LoRA 1M	teach LoRA 1M (blank)
steps	21,500	58,500	11,500	25,000
peak gradient norm (step)	34.0 (1)	70.9 (50,860)	0.441 (96)	17.4 (23,657)
mean norm: first 1% / middle / last 10%	4.80 / 1.85 / 1.39	2.69 / 4.86 / 3.77	0.183 / 0.060 / 0.056	0.169 / 5.21 / 6.80
first 1% $\div$ last 10%	$\times 3.4$ (decays)	$\times 0.7$ (grows $1.4\times$)	$\times 3.3$ (decays)	$\times 0.02$ (grows $\sim 40\times$)
cumulative gradient mass $\sum_t \ g_t\ $	44,021	378,170	783	116,877
gradient mass per step	2.05	6.46	0.068	4.68
mass share QK / VO / MLP	—	—	.18 / .55 / .27	.12 / .87 / .01

Table 6: Raw gradient strength (pre-clip global norm at every update). LoRA and full-FT norms are on different parameter sets and are compared within a method only; the LoRA pair is shown for the per-class split, which the full-FT trainer does not log.

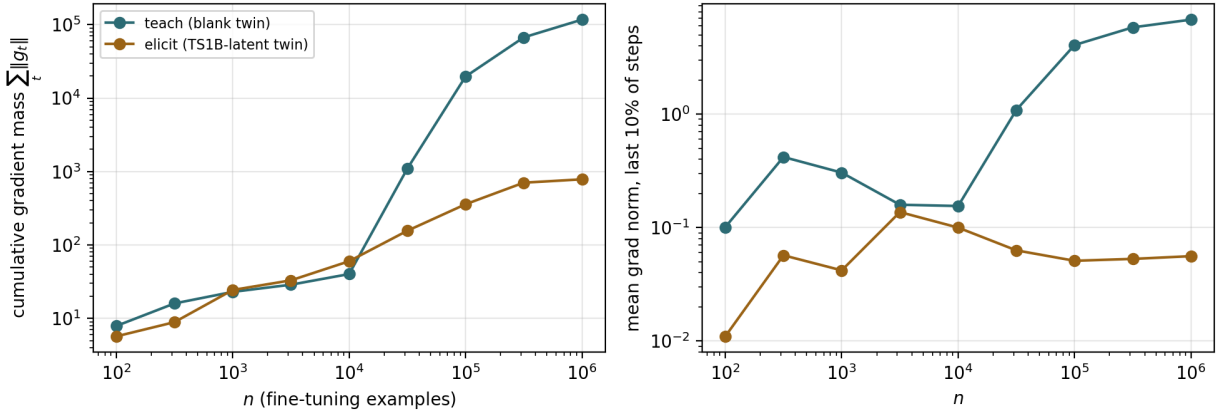


Figure 3: Cumulative gradient mass (left) and late-run gradient norm (right) vs n ; teal = teach, gold = elicit.

M7 Weight write — a small update vs. a large one

Elicit vs. teach, in one sentence. Is the change to the weights small and efficient (elicit) or large and expensive per unit of capability gained (teach)?

How it is measured. Relative weight change $\|\Delta W\|/\|W\|$ between child and parent — exact from the LoRA factors ($\Delta W = BA$), or from the checkpoint difference for full fine-tuning — over all attention and MLP matrices, with the effective rank of ΔW (participation ratio of its spectrum; 2048 = full rank). Along the LoRA runs, from the adapter snapshots: total travel $\|\Delta W_t\|_F$ vs step, its peak and final speed per step, and capability per unit travel = final exact match $\div$ total travel.

The numbers. Table 7. Relative weight write 0.050 (elicit) vs 0.086 (teach), $1.7\times$, at a comparable effective rank (554 vs 481 of 2048). The travel instrument exists only for the LoRA runs (elicit vs blank teach, 1M): write 0.095 vs 0.212, travel 120 vs 457, capability per unit travel 8.1×10^{-3} vs 2.0×10^{-4} ($40\times$). *Reading:* teaching writes about twice as much into the weights and gets forty times less capability per unit written.

	elicit	teach
relative write $\ \Delta W\ /\ W\$, full FT	0.050	0.086
effective rank of ΔW , full FT (of 2048)	554	481
relative write, LoRA 1M (teach from the blank base)	0.095	0.212
steps to convergence (LoRA 1M)	11,500	23,496
peak / final writing speed per step (LoRA 1M)	0.218 / 0.018	0.306 / 0.022
total weight travel $\ \Delta W\ _F$ (LoRA 1M)	120	457
final exact match (LoRA 1M)	0.981	0.093
capability per unit travel (LoRA 1M)	8.1×10^{-3}	2.0×10^{-4}

Table 7: Weight-space write. Top: the pair. Bottom: the LoRA runs, the only ones with snapshots for travel (curves in 05_weights/fig_weight_travel).

M8 What fine-tuning changes in the internal state — *a different change per problem vs. the same change every time*

Elicit vs. teach, in one sentence. When the child’s internal state is compared with the parent’s on the same problem, is the change different for every problem — the child has added that problem’s answer (elicit) — or is it mostly the same change regardless of the problem — the child has been moved into a new mode (teach)?

How it is measured. Take 256 problems. Run each through the parent and through the child (both in fp32) and record the residual stream at the answer position: a 2048-number vector per layer, the model’s internal state at the moment it is about to answer. Subtract: child state minus parent state. This is “the change” fine-tuning made to the internal state on that problem, one per problem. The question is whether the 256 changes look alike. *Shared-direction score:* find the single direction that best explains all 256 changes and report the fraction of their total energy that lies along it (PC1). 1.0 would mean the child moved every problem’s state by the identical amount in the identical direction; a low score means each problem’s change is its own. For calibration the same score is computed on the parent’s own 256 states (how alike they are before any change). *Control:* the score is recomputed after removing from each change the directions that spell out that problem’s answer token, so that “the same change” cannot simply be “writing the answer”. Also reported: the size of the change relative to the parent’s state, and how far the change reaches beyond the task (KL between parent and child on generic story text).

The numbers. Figure 4, Table 8. Shared-direction score of the changes at the last layer: elicit 0.056, below even the parent’s own states (0.23) — each problem’s change is its own. Teach: 0.567 — over half of every change is one and the same direction, and still 0.561 after the answer directions are removed. Control: the taught child from the blank base, which does have an output format to learn, scores 0.526, no higher; so the shared change is the algorithm’s mode, not “output a number”. The same split appears under LoRA (0.063 vs 0.528). *Reading:* elicitation adds each problem’s answer to a state that was already problem-specific; teaching first moves every problem by the same push into a new mode.

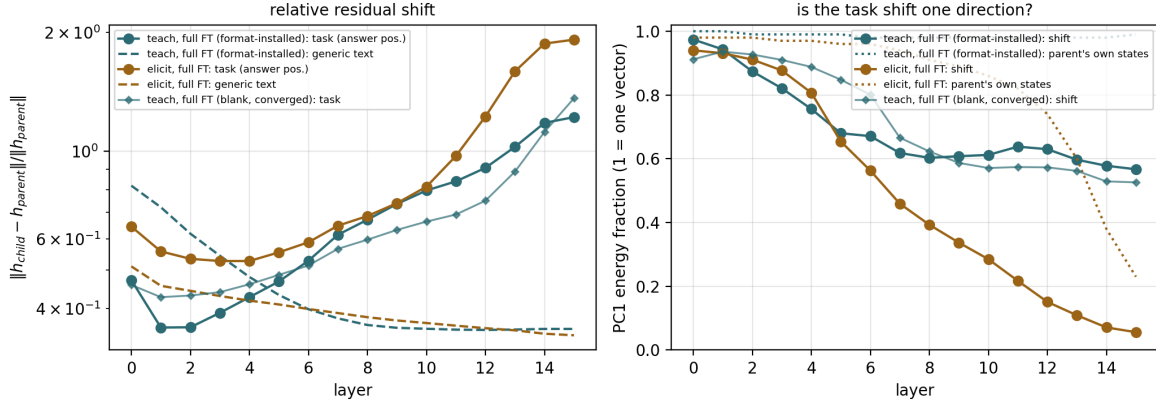


Figure 4: The pair (thick) with the blank-base taught control (thin diamonds). Left: relative residual shift by layer at the answer position (solid) and on generic story text (dashed): the size of the change by layer. Right: shared-direction score (PC1) of the 256 per-problem changes by layer, with the parent’s own states dotted.

	elicit FT	teach FT	teach FT from the blank base (control)
final exact match	0.986	0.424	0.819
shared-direction score (PC1) of the change, last layer (parent’s own states)	0.056 (0.23)	0.567 (0.99)	0.526 (0.98)
same, after removing answer-token directions	0.061	0.561	0.528
cosine of each change to the mean change	0.23	0.80	0.74
size of the change relative to the parent’s state, last layer	1.92	1.22	1.36
KL(parent child) at the answer position (nats)	16.4	12.3	22.0
KL(parent child) on stories (nats/token)	0.721	1.568	1.790

Table 8: Change of the internal state parent→child at the answer position, 256 problems, fp32. Under LoRA (1M elicit / 4M teach from the blank base): score 0.063 vs 0.528, unchanged after removing the answer directions; the LoRA-taught size is 29× (an adapter artefact).

D. The latent state: what it means and what it permits before training

M9 Latent reach — *do the words already reach the machinery?*

Elicit vs. teach, in one sentence. Before any target training, does a natural-language question already put the arithmetic components into a problem-specific state (elicit), or is there nothing for the words to reach (teach)?

How it is measured. Render each problem two ways — What is the sum of 621 and 5068? and $621 + 5068 =$ — and record every node’s activation at the answer position. Per node: the mean cosine between the two renderings of the *same* problem minus the mean cosine between renderings of *different* problems (the “bridging index”). Positive means the words put the node into the state the symbols put it in. MLP blocks only: attention heads share digit tokens across renderings and would score trivially. Compared between the symbol-only engine (no binding dose) and the elicit parent (binding dose installed). A second, geometry-only check: how problem-specific the answer-position state itself is, as PC1 and cosine-to-mean across the 256 problems (≈ 1 means the same state for every problem).

The numbers. Bridging index of the arithmetic MLPs on natural-language input: ≈ 0 in the symbol-only engine, +0.07–0.13 across layers 7–15 in the elicit parent ($10\times$). Shared-direction

score of the parent’s own answer-position states: 0.23 (elicit parent, problem-specific) vs 0.99 (teach parent; the blank base 0.98) — the same state for every problem (Table 8, “parent’s own states”). *Reading*: the words already reach the elicit parent’s machinery before any target training; the teach parents have nothing for them to reach.

M10 Answer depth — *the answer already there vs. absent; finished one layer earlier vs. later*

Elicit vs. teach, in one sentence. Is the answer readable from the parent’s internal state before it can say it (elicit) or absent (teach); and after training, is it finished one layer earlier (elicit) or later (teach)?

How it is measured. Read the model’s intermediate state at each layer as if it were the output, three ways. *Logit lens*: apply the final unembedding to the layer- ℓ state. *J-lens* (Jacobian lens): first transport the state to the final-layer basis with the average Jacobian $\partial h_{\text{final}} / \partial h_{\ell}$ over held-out story text, then unembed — this reads what the state is *disposed to make the model say*, and is where the answer can appear before the model emits it. *R-lens*: the same transport with a relevance-propagation backward pass, more faithful at early layers. Scored token = the first digit chunk of the answer (for negative answers the sign is fed in first, so predicting a minus sign never counts). Per layer: the answer token’s median rank among 128K vocabulary entries (chance 64K); the *logit-diff*, the model’s preference for the correct answer over a mismatched problem’s answer, in nats (0 = no answer-specific signal); and the *settled depth*, the first layer from which the token stays top-ranked to the end. Rank without logit-diff can be a number prior (any number ranks well in a model that emits numbers); the two are read together.

The numbers. Figure 5, Table 9. Before training: the elicit parent ranks the answer 174 of 128K in its output and 510 at layer 14 (J-lens), preferring it by 2.1 nats while it emits a rewrite; the teach parent ranks it 452 but with logit-diff 0.0 at every layer — a number, not this number (the blank base: rank $\sim$ 75K, logit-diff 0.0). After training: the elicited child settles at layer 14 (J-rank 338 / 8 / 1 at layers 12 / 13 / 14), the trajectory of the parent’s engine on symbols (122 / 6 / 1); the taught child settles at layer 15 (27,044 / 15,058 / 961), as does the taught child from the blank base. *Reading*: the elicit parent already carries the answer and elicitation amplifies the read-out; the teach parent carries nothing, and a taught model finishes the answer one layer later even when it solves the task.

model	surface	own acc	rank L12 (logit / J / R)	rank L14 (logit / J / R)	read-out	ld
elicit parent	NL	0.047	5,704 / 2,606 / 2,730	436 / 510 / 424	174	+2.1
teach parent (format-installed)	NL	0.004	8,394 / 4,036 / 5,417	1,106 / 997 / 998	452	0.0
elicited child (FT)	NL	0.988	1,797 / 338 / 492	1 / 1 / 1	1	+18.7
taught child (FT)	NL	0.535	23,909 / 27,044 / 25,529	381 / 961 / 1,848	1	+14.2
blank base	NL	0.000	76,049 / 80,840 / 78,243	77,707 / 77,320 / 79,116	75,441	0.0
taught child from the blank base (FT)	NL	0.844	18,988 / 4,123 / 9,620	170 / 138 / 459	1	+16.1
elicit parent (the engine)	symbols	0.836	1,224 / 122 / 334	1 / 1 / 1	1	+15.1

Table 9: Median rank of the answer’s first digit token among 128K entries under the three lenses; “own acc” = the model’s own first-digit-token accuracy on the 256 problems; read-out = the last layer, where all lenses equal the model’s output; ld = logit-diff at the read-out (nats). Bottom: the blank base (rank without a number prior), the taught child from it (settles at layer 15 too), and the engine’s own trajectory on symbols, which the elicited child reproduces.

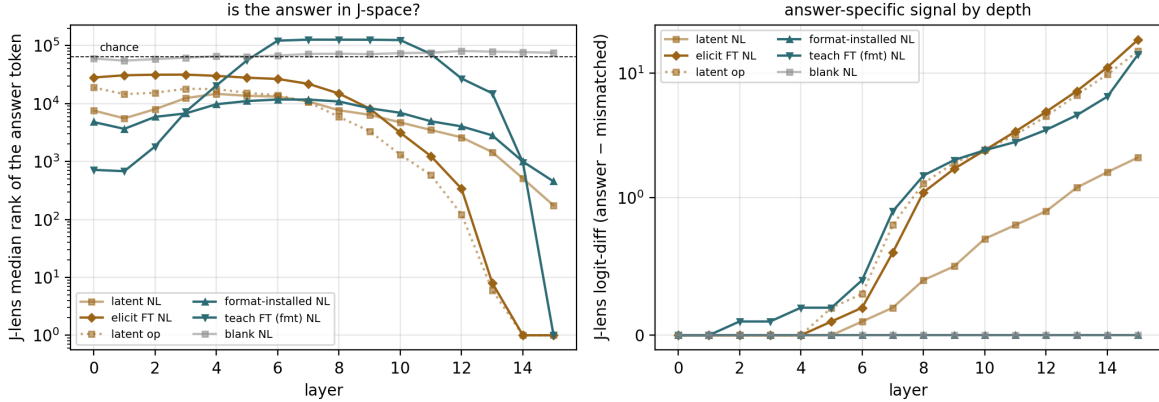


Figure 5: J-lens read-out of the answer token by layer: median rank (left, chance 64K) and logit-diff against a mismatched problem’s answer (right). Gold = elicit (parent: squares; child: diamonds; dotted: the parent on the symbol surface), teal = teach (parent: up-triangles; child: down-triangles), grey = blank base.

M11 Switch-on by patching — can the capability be turned on without training?

Elicit vs. teach, in one sentence. With the parent’s weights frozen, does writing the fine-tuned child’s internal activations into the parent make it solve the task — possible only if the machinery is already there (elicit) and impossible if it is not (teach)?

How it is measured. Donor patching. Take the fine-tuned child (the donor) and its parent (the patient). For each problem, run the child and record its activations at the 32 circuit nodes of M2 at the answer position; run the parent on the same problem and overwrite those 32 activations with the child’s; let the parent finish. Two variants: *per-prompt states* (each problem gets the child’s activations on that problem) and *mean vector* (every problem gets the child’s average activation, a single constant shift). Control: the same with 32 random nodes. Read-outs: exact match, and “format” = the fraction of outputs that are a number at all. Done for the elicit side (elicited child into the elicit parent) and the teach side (each taught child into its own parent: the blank base, or the format-installed base).

The numbers. Table 10, Figure 6. Elicit parent: the child’s per-prompt states patched into 32 nodes give exact match 0.121 from a parent that scores 0.000 (random nodes 0.000; the mean vector changes only the output format). Teach parent: its child’s states patched in give 0.016 (into the blank base, from the blank-base child: 0.004). *Reading:* 32 activations from the child are enough to switch the latent capability on because the machinery underneath is the parent’s own; the same patch from a taught child finds nothing underneath in its parent.

E. Every metric on one page

The pair, one row per metric; the third column is the taught twin fine-tuned from the blank base with the same recipe, the robustness check: it performs better (0.819) and agrees with the format-installed taught twin on every row.

Patient + donor (32 circuit nodes)	condition	format	exact match
elicit parent + elicited child’s per-prompt states	circuit nodes	0.387	0.121
teach parent + taught child’s per-prompt states	circuit nodes	0.836 (parent 1.000)	0.016
elicit parent + LoRA child’s per-prompt states	circuit nodes	0.359	0.125
	random-32 nodes	0.051	0.000
elicit parent + LoRA child’s mean vector	circuit nodes	0.164	0.000
blank base + blank-base taught child’s states	every condition	≤ 0.02	≤ 0.004

Table 10: Donor patching, weights frozen. The parents score 0.000 unpatched. Bottom: controls (random nodes, the mean vector, the LoRA child as donor, the blank-base pair).

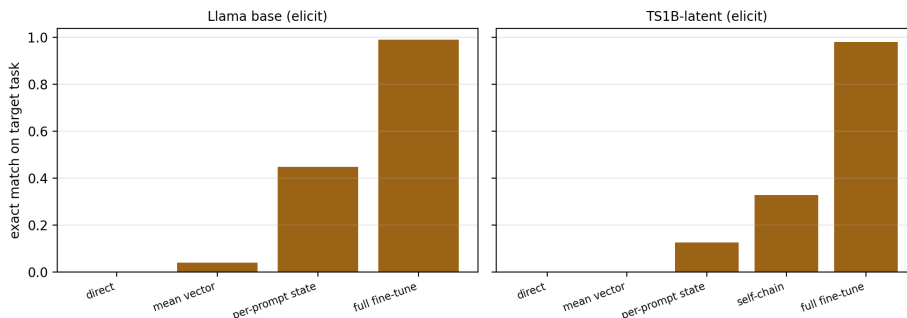


Figure 6: Donor patching on the elicit parent (right; left: the same on a Llama-3.2-1B parent, for intuition), weights frozen except the last bar: direct, mean vector, per-prompt states, and the fine-tuned child for scale; the self-chain bar is the construction check of Table 1, not a patch.

Notes on measurement

- **Teach-side metrics are measured on a taught model that solves the task.** The first taught endpoint (LoRA 1M, exact match 0.093) had not learned it. Every teach-side metric was re-run on fully fine-tuned taught endpoints (from the format-installed and the blank base), with the fully fine-tuned elicited endpoint as the method control. Three earlier readings changed and are stated in their revised form above: taught operand roles are mostly shared rather than absent (M4), the circuit difference is partial rather than total (M2), and the size of the taught state change follows the fine-tuning method (M8).
- **Answer depth is scored on the first digit token.** Scoring the raw first token credited a taught child with layer-5 formation on 48 problems — all subtractions with a negative answer, whose first token is the minus sign, produced by a comparison circuit. With the sign fed in first the effect vanishes; the other models were unaffected.
- **Construction controls.** The same doses on the blank twin change nothing on the target (0.000 in every probe); a sum-only binding dose leaves “difference” unbound; the composition law self-chain = rewrite $\times$ compute held on four variants.

	elicit FT	teach FT	teach FT from the blank base	metric
exact match (0-shot)	0.986	0.424	0.819	M1
steps to stop (passes)	21,500 (0.7)	58,500 (1.9)	72,000 (2.3)	M1
held-out test loss (nats/token)	0.015	0.639	0.154	M1
circuit overlap with the elicited circuit	–	0.306	0.333	M2
top-8 nodes sufficient / necessary	0.923 / 0.943	0.983 / 0.988	0.950 / 0.965	M3
operand-reading heads vs the engine's (first / second)	0.694 / 0.857	0.879 / 0.659	0.763 / 0.658	M4
gradient norm, first 1% → last 10% of steps	4.80 → 1.39 (decays 3.4×)	2.69 → 3.77 (grows 1.4×)	2.30 → 3.43 (grows 1.5×)	M6
cumulative gradient mass	44,021	378,170	353,277 (+ 30,954)	M6
relative weight write $\ \Delta W\ /\ W\ $	0.050	0.086	0.102	M7
shared-direction score of the state change (after removing answer content)	0.056 (0.061)	0.567 (0.561)	0.526 (0.528)	M8
answer settled depth, median (quartiles)	layer 14 (14–14)	layer 15 (15–15)	layer 15 (15–15)	M10
J-space rank of the answer at layers 12 / 13 / 14	338 / 8 / 1	27,044 / 15,058 / 961	4,123 / 1,832 / 138	M10
child's states patched into its own parent, per-prompt (mean)	0.121 (format only)	0.016 (0.000)	0.004 (0.000)	M11

Table 11: The pair and the blank-base robustness check, one row per metric.